



Functional Requirements Document

Commercial Profitability Executive Dashboard

1. Problem Statement

Orbita Digital Group is a global distributor of consumer electronics and home goods, operating in 36 countries across six continents (North America, South America, Europe, Asia, Africa, and Oceania) through a portfolio of seven major brands (Sony, LG, Philips, Sonos, Microsoft, Ninja, and Kid Toys) spanning 20 product categories and 444 unique SKUs.

Over the past several fiscal periods, the commercial team has relied on a set of disconnected, region-specific reports to track total revenue by market, brand, and product line. While these reports provide a reasonably accurate picture of sales volume, they do not capture the underlying cost structure of each transaction, and therefore offer no visibility into gross profitability at any level of granularity - whether overall, by country, by brand, by category, or by individual product.

This gap in visibility has had a measurable and undesirable effect on commercial decision-making. Because performance has historically been evaluated primarily through revenue figures, the sales and account management teams have gravitated toward promoting and prioritizing high-revenue products and brands, some of which may carry materially thinner margins than lower-revenue alternatives. In several informally observed instances, products responsible for a disproportionate share of total revenue have also been suspected of having weaker unit economics than other, lower-volume products - though without a centralized and reliable source of truth, these observations have never been formally validated, quantified, or acted upon.

Compounding this issue is the fact that the company's supplier negotiation strategy is currently informed almost exclusively by purchase volume and revenue contribution per brand, rather than by the realized profitability those brands generate once cost of goods sold is accounted for. This creates a structural risk: the company may be allocating disproportionate negotiating leverage, marketing investment, and inventory commitment toward brands and product lines that, while high-performing in gross revenue terms, do not deliver a commensurate return in gross profit.

The Commercial Direction has therefore requested a centralized, single-source-of-truth executive dashboard that reconciles sales volume with gross profitability across every dimension currently tracked informally - continent, country, brand, category, product, and time period - so that commercial strategy, supplier negotiations, and regional resource allocation can be grounded in actual profitability rather than revenue alone. The dashboard must make explicit, at a glance, any divergence between what sells the most and what earns the most, since this exact blind spot is the root cause of the inefficiencies described above.

2. Gross Profit Margin Target

The Commercial Direction has established an internal benchmark of a **35% Gross Profit Margin** as the minimum acceptable threshold for the overall commercial operation. This target must be used as the reference point against which the dashboard's overall Gross Profit Margin, as well as the Gross Profit Margin broken down by brand and by category, is evaluated. Any segment performing below the 35% threshold must be visually flagged with an alert indicator, while segments meeting or exceeding the target should be flagged with a positive indicator.

3. General Objective

To design an executive dashboard in Power BI, supported by an exploratory data analysis in Python (data cleaning, null detection, and unique value identification), that visualizes Total Revenue, Total Cost of Sale, and Gross Profit across the commercial operation - enabling the identification of the markets, brands, and categories that are most profitable in Gross Margin terms, not merely those with the highest sales volume.

4. Specific Objectives

- Display Total Revenue and Total Cost of Sale for the period.
- Calculate and display Total Gross Profit and the overall Gross Profit Margin (%).
- Display the product with the highest Revenue and the product with the highest Gross Profit Margin (these may not be the same product).
- Visualize sales by continent.
- Display the top 5 countries by Revenue.
- Display historical sales revenue and number of transactions by month.
- Display Revenue and Gross Profit Margin by brand.
- Display the category with the highest Revenue and the category with the highest Gross Profit Margin.
- Display sales by year, with a year-over-year comparison.
- Calculate the Average Ticket (Revenue per transaction).
- Compare Gross Profit Margin (overall, by brand, and by category) against the 35% reference target defined by the Commercial Direction.

5. Dataset Description

The file "Ventas_Tech.xlsx" contains the historical sales record of Orbita Digital Group between June 2020 and August 2022. Each row represents an individual sales transaction, including product, brand, category, location (country and continent combined in a single text field), price, cost, and quantity sold.

Metric	Value
Number of records	22,419 (10 fully empty rows to be removed)
Period covered	06/2020 - 08/2022
Unique countries	36
Brands	7 (Sony, LG, Philips, Sonos, Microsoft, Ninja, Kid Toys)
Categories	20
Unique products	444

Data Dictionary

Field	Type	Description
Fecha De Venta	Date	Date the sale took place
Ubicación	Text	Country and continent combined, e.g. 'China, Asia' (requires splitting)
Producto	Text	Name/model of the product sold
Marca	Text (category)	Product brand
Categoría	Text (category)	Product category
Precio Unidad	Decimal	Unit sale price
Costo Unidad	Decimal	Unit cost
Cantidad Vendida	Integer	Units sold in the transaction
Facturación	Decimal	Revenue = Precio Unidad x Cantidad Vendida

Note: Gross Profit, Gross Profit Margin (%), Total Cost of Sale, and the Country/Continent split are derived from these base fields through Python-based EDA, Power Query transformations, and Power BI quick measures.